

THE STRUCTURE OF MATHEMATICAL REALITY

Phase IV — Chapter 14

What a Model Is and Why It Works

Note Template

Self-Study Curriculum

How to use this template

Complete these notes **after** the reading and the lecture. The goal is not to copy phrases from the text — it is to reconstruct the ideas in your own language. Leave every item blank on first pass; fill it in from memory, then check.

1. A model is a _____ from a _____ to a _____.

2. A model works when the correspondence is

the technical term and what it means

3. “The map is not the territory” misses what

what the phrase gets right and what it gets wrong

4. Three modes of model failure

a. Structural misspecification:

b. Resolution failure:

c. Boundary failure:

For each, which concept from the curriculum diagnoses it:

a.

b.

c.

5. Bayesian Occam's razor

Why simpler models are *more strongly supported* (not just more convenient):

6. Connection to generative equations

Why a generative equation is stronger evidence than a statistical summary:

7. An example of each failure mode from my own work

- a.
- b.
- c.

8. My revised answer to “what happens when a model works?”

compare to your answer from Chapter 6
